

CAPE FEAR REGIONAL, NC, USA

WMO: 722191

Lat: 33.929N

Lon: 78.075W

Elev: 25

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03721

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.6	28.1	4.6	7.0	29.3	10.2	9.2	33.3	20.3	45.7	18.7	46.8	5.5	10	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	89.8	78.3	87.9	77.6	86.3	76.8	80.5	86.2	79.5	85.3	78.4	84.0	8.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	150.6	84.0	77.5	143.0	83.1	76.9	140.1	82.7	44.0	86.2	43.0	85.3	42.0	83.8	97.7

Extreme Annual Design Conditions

Extreme Annual WS							Extreme Annual Temperature										n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years										
Min	Max				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max										
(a)	(b)	(c)			(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)										
18.3	16.2	14.0			17.1	93.3	6.3	4.4	12.5	96.5	8.8	99.0	5.3	101.5	0.7	104.7										
			DB		15.5	83.2	2.7	4.5	13.6	86.4	12.0	89.0	10.5	91.6	8.6	94.9										
			WB																							

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	45.4	48.5	54.2	62.7	70.5	77.5	79.6	78.4	74.8	65.7	55.4	50.3	(6)
	DBStd	13.99	9.97	8.32	8.76	6.73	6.04	4.37	3.66	4.72	4.40	7.83	8.17	9.12	(7)
	HDD50	545	212	118	59	3	0	0	0	1	0	4	36	112	(8)
	HDD65	2417	607	462	347	118	22	1	0	5	1	89	303	462	(9)
	CDD50	5531	70	77	189	384	636	824	917	883	744	490	197	120	(10)
	CDD65	1928	0	1	12	49	194	374	452	422	295	110	14	5	(11)
	CDH74	15651	0	2	47	119	1045	3265	4483	3933	2159	564	32	2	(12)
	CDH80	3967	0	0	12	6	159	805	1364	1133	435	53	0	0	(13)
(14) Wind	WSAvg	6.3	6.2	6.5	6.9	7.1	6.6	6.7	6.6	5.6	5.9	5.8	5.8	5.8	(14)
(15) Precipitation	PrecAvg	56.4	4.2	3.3	4.1	2.8	3.8	4.3	5.9	7.2	7.9	4.7	3.3	3.7	(15)
	PrecMax	83.6	10.3	10.8	9.5	7.2	10.0	11.5	15.0	14.2	28.3	10.6	8.0	8.4	(16)
	PrecMin	28.3	0.7	1.4	1.3	0.3	1.2	0.7	2.2	1.8	1.9	0.2	0.6	0.9	(17)
	PrecStd	12.5	2.4	2.0	2.0	1.9	2.1	2.4	3.0	3.3	6.1	3.0	1.9	2.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.9	72.8	79.4	80.4	90.9	91.0	91.3	91.0	88.3	84.2	77.6	73.0	(19)
		MCWB	63.4	65.1	69.4	69.4	82.2	77.9	79.1	78.1	76.0	74.7	70.0	69.2	(20)
	2%	DB	66.0	67.7	73.0	77.2	83.3	88.0	89.7	88.5	86.2	81.4	74.0	70.0	(21)
		MCWB	62.6	61.5	64.2	66.6	73.9	77.2	78.6	78.1	75.5	72.3	67.8	66.4	(22)
	5%	DB	63.6	64.3	70.2	74.8	81.2	86.1	87.9	87.5	84.4	79.3	71.6	67.1	(23)
		MCWB	60.3	59.3	63.6	66.4	72.2	76.6	77.8	77.7	74.9	71.3	66.4	63.7	(24)
	10%	DB	61.0	61.7	67.2	72.7	79.4	84.4	86.2	85.8	82.3	77.3	68.8	64.2	(25)
		MCWB	57.7	56.6	61.0	64.9	70.9	75.6	77.1	76.7	73.3	70.0	64.2	61.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.7	67.3	72.9	74.1	84.9	81.4	81.5	81.8	79.2	77.3	72.6	70.5	(27)
		MCDB	67.3	71.0	77.8	77.8	92.9	87.5	88.1	86.5	84.6	80.9	76.6	72.4	(28)
	2%	WB	63.1	64.5	67.0	70.4	76.3	79.5	80.3	80.2	77.9	75.7	69.9	67.4	(29)
		MCDB	65.0	66.4	71.4	74.6	80.9	85.1	86.7	85.5	83.1	79.6	72.4	68.8	(30)
	5%	WB	61.0	60.8	64.5	68.5	74.4	78.3	79.4	79.1	76.7	73.6	67.6	64.2	(31)
		MCDB	63.4	63.7	68.4	72.5	79.2	83.6	85.6	84.7	82.1	77.9	70.4	66.0	(32)
	10%	WB	58.5	57.9	62.4	66.8	72.9	77.2	78.3	78.0	75.4	71.8	64.8	61.8	(33)
		MCDB	61.0	60.7	66.0	71.1	77.8	82.4	84.3	83.7	81.1	76.2	68.0	64.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	16.6	16.8	16.0	14.1	12.3	12.4	13.0	13.9	16.4	18.1	16.6	(35)
		MCDBR	16.8	17.8	17.5	16.3	14.3	13.2	13.1	13.0	14.4	15.3	17.5	15.6	(36)
		MCWBR	14.0	12.6	10.7	9.5	7.6	6.7	6.1	6.1	6.6	8.6	12.6	12.2	(37)
	5% WB	MCDBR	15.2	16.4	15.3	13.2	12.5	11.9	11.4	11.5	12.0	11.9	15.1	13.9	(38)
		MCWBR	13.6	13.4	11.0	8.9	7.6	7.1	6.3	6.1	6.4	7.7	12.9	12.2	(39)
(40) Clear-Sky Solar Irradiance	taub		0.331	0.339	0.367	0.399	0.456	0.484	0.521	0.499	0.428	0.387	0.351	0.335	(40)
	taud		2.517	2.493	2.436	2.333	2.228	2.199	2.109	2.167	2.360	2.445	2.496	2.534	(41)
	Ebn at Noon		279	289	288	282	266	257	247	251	265	268	269	271	(42)
	Edh at Noon		27	30	35	40	45	46	50	47	37	31	27	25	(43)
(44) All-Sky Solar Radiation	RadAvg		856	1078	1439	1824	2004	2068	1953	1755	1476	1249	952	760	(44)
	RadStd		56	104	125	123	132	125	123	106	109	129	70	56	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (41 neighbors)		+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+297	+214	

Nomenclature: See separate page